Lecture 22

Transforms of Discontinuous Functions and Powers

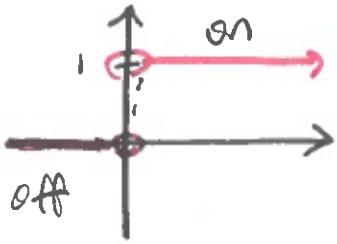
Now that we have a foundational knowledge of Laplace transforms and their use in solving IVPs, we can turn our attention to studying the motivating ideas behind the usage of transforms: solving differential equations involving functions that have jump discontinuities.

We have seen that we can still determine the Laplace transform of a piecewise continuous function that may have jump discontinuities by separating the definition into a collection of integrals, but we do not yet have the understanding of how to find the inverse transform of a discontinuous function.

To achieve this, we begin with the following definition:

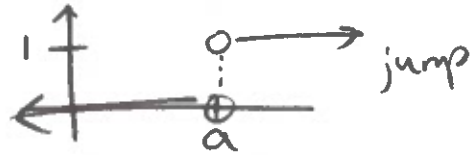
Definition (Heaviside's Unit Step Function)**:** The **unit step function**, $u(t)$, is defined by

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$



switch

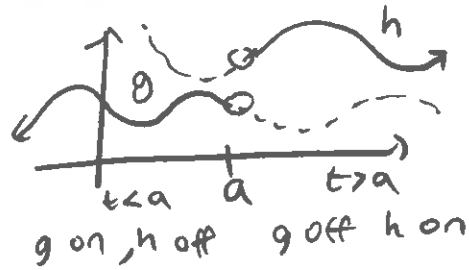
shift



Note: $u(t-a) = \begin{cases} 0, & t-a < 0 \\ 1, & t-a > 0 \end{cases} = \begin{cases} 0, & t<a \\ 1, & t>a \end{cases}$ and $M \cdot u(t-a) = \begin{cases} 0, & t<a \\ M, & t>a \end{cases}$

We can use the unit step function to rewrite piecewise-defined functions as single expressions. Consider

$$f(t) = \begin{cases} g(t), & t < a \\ h(t), & t > a \end{cases}$$



Then $f(t) = g(t) - g(t)u(t-a) + h(t)u(t-a) = g(t) + u(t-a)(h(t)-g(t))$

off g

on h

Example: Consider $f(t) = \begin{cases} 10, & t < 5 \\ t-5, & t > 5 \end{cases}$

$f(t) = 10 + u(t-5)(t-5 - 10) = 10 + u(t-5)(t-15)$

on off

In order for the unit step function to be useful with Laplace transforms, we will need to determine its transform.

Theorem (Transform of the Unit Step Function): For $a \geq 0$, we have

$$\mathscr{L}\{u(t-a)\} = \frac{e^{-as}}{s}, \quad s > 0$$

Proof: $\mathscr{L}\{u(t-a)\} = \int_0^\infty e^{-st} u(t-a)\,dt = \int_0^a e^{-st} \underbrace{u(t-a)}_{0}\,dt + \int_a^\infty e^{-st} \underbrace{u(t-a)}_{1}\,dt$

$\begin{cases} 0, & t<a \\ 1, & t>a \end{cases}$ $t<a$ $t>a$

$= 0 + \int_a^\infty e^{-st} \cdot 1\,dt = \lim_{N\to\infty} \int_a^N e^{-st}\,dt = \lim_{N\to\infty} -\frac{1}{s} e^{-st} \Big|_{t=a}^{t=N}$

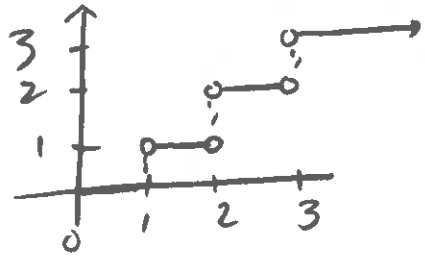
$= \lim_{N\to\infty} \left(-\frac{1}{s} \cancel{e^{-sN}}^{\,0} - \left(-\frac{1}{s} e^{-s\cdot a}\right)\right) = 0 + \frac{e^{-as}}{s} = \boxed{\frac{e^{-as}}{s}}$ $\square$

$s>0$

Note $\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}$

Example: Determine the Laplace transform of

$$f(t) = \begin{cases} 0, & 0 < t < 1 \\ 1, & 1 < t < 2 \\ 2, & 2 < t < 3 \\ 3, & t > 3 \end{cases}$$



$$f(t) = 0 \cdot u(t) + u(t-1)(1-0) + u(t-2)(2-1) + u(t-3)(3-2)$$
$$= u(t-1) + u(t-2) + u(t-3)$$

So $\mathcal{L}\{f(t)\} = \mathcal{L}\{u(t-1) + u(t-2) + u(t-3)\}$

$$= \frac{e^{-s}}{s} + \frac{e^{-2s}}{s} + \frac{e^{-3s}}{s}$$

$$e^{-3s} \cdot \frac{1}{s}$$

As $\mathcal{L}\{u(t-a)\} = \dfrac{e^{-as}}{s}$, we have that $\mathcal{L}^{-1}\left\{\dfrac{e^{-as}}{s}\right\} = u(t-a)$.

We have previously seen that multiplication by an exponential prior to taking the transform shifts the resulting transform. How does multiplying by an exponential prior to taking an inverse transform affect the resulting function?

Theorem (Translation in t): Suppose $\mathscr{L}\{f(t)\} = F(s)$ exists for $s > \alpha \geq 0$. If $a > 0$, then

$$\mathscr{L}\{f(t-a)u(t-a)\} = e^{-as}F(s).$$

$\mathscr{L}\{u(t-a)\}$

$\mathscr{L}\{1 \cdot u(t-a)\} = \frac{e^{-as}}{s}$

$= e^{-as} \cdot \frac{1}{s}$

$\mathscr{L}\{1\}$

or conversely,

$$\mathscr{L}^{-1}\{e^{-as}F(s)\} = f(t-a)u(t-a)$$

Proof:

$$\mathscr{L}\{f(t-a)u(t-a)\} = \int_0^\infty e^{-st} f(t-a)u(t-a)\,dt$$

$$= \underbrace{\int_0^a e^{-st} f(t-a)\cancel{u(t-a)}^{\,0}\,dt}_{t<a} + \underbrace{\int_a^\infty e^{-st} f(t-a)\cancel{u(t-a)}^{\,1}\,dt}_{t>a}$$

$$= \int_a^\infty e^{-st} f(t-a)\,dt$$

(*) use $w = t - a$

$\Rightarrow t = w + a$

$dw = dt$

$t = a \Rightarrow w = a - a = 0$

$t \to \infty,\ w \to \infty$

not 0

$$= \int_0^\infty e^{-s(w+a)} f(w)\,dw$$

$$= \int_0^\infty e^{-sw} e^{-sa} f(w)\,dw = e^{-as} \underbrace{\int_0^\infty e^{-sw} f(w)\,dw}_{\mathscr{L}\{f\}} = e^{-as}F(s) \quad \square$$

Note: If $f(t) = 1$, then

$$\mathscr{L}\{f(t-a)u(t-a)\} = e^{-as}F(s) = e^{-as} \cdot \frac{1}{s}.$$

Also, if $g(t) = f(t-a)$, then $f(t) = g(t+a)$ and

$$\mathscr{L}\{g(t)u(t-a)\} = \mathscr{L}\{f(t-a)u(t-a)\} = e^{-as}\,\mathscr{L}\{f(t)\} = e^{-as}\,\mathscr{L}\{g(t+a)\}$$

$$\mathscr{L}\{g(t)u(t-a)\} = e^{-as}\mathscr{L}\{g(t+a)\}$$

$g(t) = t^3$

Example: Determine $\mathcal{L}\{t^3 u(t-1)\} = e^{-1\cdot s}\mathcal{L}\{g(t+1)\}$ (a)

$$= e^{-s}\mathcal{L}\{(t+1)^3\} = e^{-s}\mathcal{L}\{t^3+3t^2+3t+1\}$$

$$= e^{-s}\left(\frac{6}{s^4} + 3\cdot\frac{2}{s^3} + 3\cdot\frac{1}{s^2} + \frac{1}{s}\right)$$

Example: Determine $\mathcal{L}^{-1}\left\{\frac{e^{-3s}}{s^2+9}\right\} = \mathcal{L}^{-1}\left\{e^{-3s}\cdot\frac{1}{s^2+9}\right\}$ (a)

$\frac{1}{3}\cdot\frac{3}{s^2+9}$ $\sin(3t)$

$$\boxed{\mathcal{L}^{-1}\{e^{-as}F(s)\} = f(t-a)u(t-a)}$$

$$= \left(\frac{1}{3}\sin(3t)\right)\Big|_{t\to t-3}\cdot u(t-3)$$

$$\boxed{\mathcal{L}^{-1}\left\{\frac{e^{-3s}}{s^2+9}\right\} = \frac{1}{3}\sin(3(t-3))\,u(t-3)}$$

Example: Solve $y'' + y = u(t-\pi)$, $y(0) = 1$, $y'(0) = 0$

+1 at $t=\pi$

$$\mathcal{L}\{y''+y = u(t-\pi)\} \Rightarrow (s^2Y(s) - s\cdot 1 - 0) + Y(s) = \frac{e^{-\pi s}}{s}$$

(a)

$$\Rightarrow Y(s)(s^2+1) - s = \frac{e^{-\pi s}}{s}$$

$$\Rightarrow Y(s) = \frac{\frac{e^{-\pi s}}{s} + s}{s^2+1} = e^{-\pi s}\cdot\frac{1}{s(s^2+1)} + \frac{s}{s^2+1}$$

PFD: $\frac{1}{s(s^2+1)} = \frac{A_1}{s} + \frac{A_2 s + B_2}{s^2+1}$

$\Rightarrow A_1 = 1,\ A_2 = 1,\ B_2 = 0$

$\Rightarrow \frac{1}{s} - \frac{s}{s^2+1}$

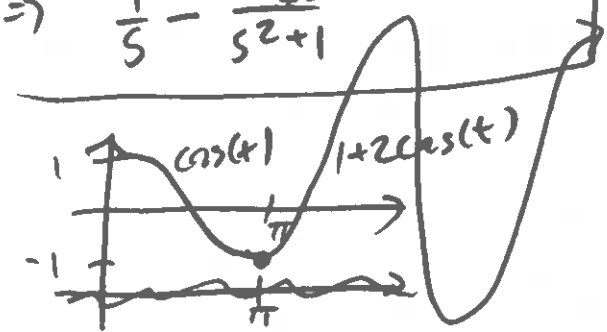
$$\Rightarrow Y(s) = e^{-\pi s}\left(\frac{1}{s} - \frac{s}{s^2+1}\right) + \frac{s}{s^2+1}$$

$$y(t) = \underbrace{(1-\cos(t))\Big|_{t\to t-\pi}}_{\text{particular soln}}\cdot u(t-\pi) + \underbrace{\cos(t)}_{\text{homo soln.}}$$

$$= (1-\cos(t-\pi))\,u(t-\pi) + \cos(t)$$

$$= (1+\cos(t))\,u(t-\pi) + \cos(t)$$

$$= \begin{cases} \cos(t), & t<\pi \\ 1+2\cos(t), & t>\pi \end{cases}$$



C-E can have $y_h = c_1 t^r + c_2 t^{\ell}$

So far we have been able to find the transform of various functions including those which are solutions to homogeneous constant coefficient linear equations. We have seen that certain variable coefficient equations, such as Cauchy-Euler equations, have powers as their solutions. We have only attempted to find the transform of powers whose exponent is a positive integer, i.e. $\mathscr{L}\{t^n\} = \frac{n!}{s^{n+1}}$.

Ex: $\mathscr{L}\{\sqrt{t}\}$?

What if the exponent is a real number?

Consider $f(t) = t^r$ with $r > -1$ (why?). We have

$$\mathscr{L}\{t^r\} = \int_0^\infty e^{-st} t^r \, dt$$

$t = \frac{1}{s}u$

By making the substitution $u = st$, then $dt = \frac{1}{s}du$ and $t = \frac{u}{s}$ and so

$$\mathscr{L}\{t^r\} = \int_0^\infty e^{-st} t^r \, dt = \int_0^\infty e^{-u} \underbrace{\left(\frac{u}{s}\right)^r}_{t} \underbrace{\left(\frac{1}{s}\right)}_{dt} du = \frac{1}{s^{r+1}} \underbrace{\int_0^\infty e^{-u} u^r \, du}_{\text{number if it converges}}$$

If we compare this result with $\mathscr{L}\{t^n\} = \frac{n!}{s^{n+1}}$, we can see that the difference between the two results is the power of s is multiplied by an integral instead of a factorial. We make the following definition to unify the two results:

Definition (The Gamma Function)**:** The **gamma function**, $\Gamma(r)$, is defined by

$$\Gamma(r) = \int_0^\infty e^{-u} u^{r-1} \, du, \quad r > 0$$

Note: We have u^{r-1} in the definition to align with certain properties. The integral we found before had u^r which would correspond to $\Gamma(r+1)$. It can be shown that the gamma function converges for $r > 0$.

$\Gamma(r) = \int_0^\infty e^{-u} u^{r-1} du$? $= \frac{1}{s^{r+1}} \underbrace{\int_0^\infty e^{-u} u^r du}_{\Gamma(r+1)}$

$\mathcal{L}\{t\} = \frac{\Gamma(2)}{s^2} = \frac{1!}{s^2}$

With this we can now say that $\mathscr{L}\{t^r\} = \frac{\Gamma(r+1)}{s^{r+1}}$ for $r > 0$, but we have a situation where there are two conflicting results about the transform of a power. Are the gamma function $\Gamma(r)$ and the factorial $n!$ related? ?

One property of factorials is that $\boxed{(n+1)! = (n+1) \cdot n!}$. There is a similar property for the gamma function. Note,

$$\Gamma(r+1) = \int_0^\infty e^{-u} u^r \, du = \lim_{N\to\infty} \int_0^N e^{-u} u^r du$$

IBP:

$u^r \searrow + \; e^{-u}$

$ru^{r-1} \xrightarrow{\int} -e^{-u}$

$$= \lim_{N\to\infty} \left[-e^{-u} u^r \Big|_{u=0}^{u=N} + \int_0^N e^{-u} r u^{r-1} du \right]$$

$$= \lim_{N\to\infty} \left[-\cancel{e^{-N}} \cdot N^r + \cancel{e^0} \cdot 0^r + r \int_0^N e^{-u} u^{r-1} du \right]$$

$$= r \cdot \int_0^\infty e^{-u} u^{r-1} du = r\Gamma(r)$$

$\Gamma(n+1) = n\Gamma(n) = n(n-1)\Gamma(n-1) = \cdots$ and $\Gamma(1) = 1$

Thus, the definition of the gamma function aligns with the factorial as $\Gamma(n+1) = n!$ for n a positive integer.

$\Gamma(n+1) = n!$ n pos integer.

Example: Given that $\Gamma(1/2) = \sqrt{\pi}$ (why?) evaluate $\mathscr{L}\{t^{5/2} e^{4t}\}$

$\boxed{\mathcal{L}\{t^r\} = \frac{\Gamma(r+1)}{s^{r+1}}}$

$\boxed{\mathcal{L}\{e^{at} f(t)\} = \mathcal{L}\{f(t)\}\Big|_{s\to s-a}}$

$\mathcal{L}\{t^{5/2} e^{4t}\} = \mathcal{L}\{t^{5/2}\}\Big|_{s\to s-4}$ (r)

$= \frac{\Gamma(\frac{5}{2}+1)}{s^{5/2+1}}\Big|_{s\to s-4}$

$\Gamma(r+1) = r\Gamma(r)$

$= \frac{\Gamma(\frac{7}{2})}{(s-4)^{7/2}} = \frac{\frac{5}{2}\Gamma(\frac{5}{2})}{(s-4)^{7/2}} = \frac{\frac{5}{2}(\frac{3}{2})\Gamma(\frac{3}{2})}{(s-4)^{7/2}} = \frac{\frac{5}{2}(\frac{3}{2})(\frac{1}{2})\Gamma(\frac{1}{2})}{(s-4)^{7/2}}$

$= \boxed{\frac{15\sqrt{\pi}}{8(s-4)^{7/2}}}$